Efficient Auditory Coding

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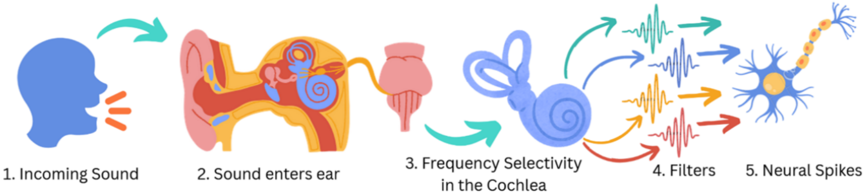
Agenda

- Getting to know each other
- A refresher: what is the project about?
- Research questions
- Overview of timeline
- What to do till week 1?
- Set meetings

Introduction & Motivation

- Human speech communication is robust in a wide range of conditions
  - Context
  - Redundancy in the waveform
  - More options?
- A better understanding of why this is the case can help improve speech technology
- Starting point: the *efficient coding hypothesis*(Barlow 1961)[1] (for speech)



[1] H. B. Barlow (Sept. 1961). "Possible Principles Underlying the Transformations of Sensory Messages". In: *Sensory Communication*. The MIT Press, pp. 216–234. DOI: 10.7551/mitpress/9780262518420.003.0013

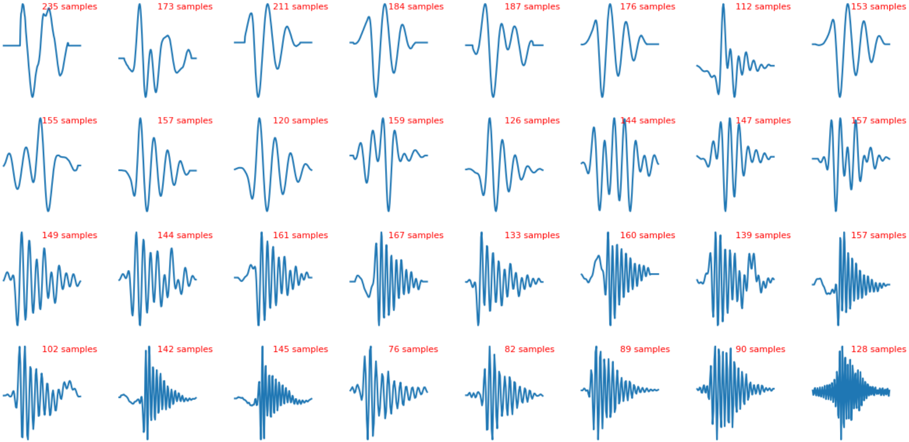
Background (1): Smith & Lewicki

- In 2006, Smith and Lewicki(Smith and Lewicki 2006)[2] tried to find such an efficient representation.
- Idea: train a dictionary of $N_g = 32$ kernels and use these to represent the waveform in a sparse manner.
- This was done using matching pursuit.

[2] Evan C. Smith and Michael S. Lewicki (Feb. 2006). "Efficient auditory coding". In: *Nature* 439.7079, pp. 978–982. ISSN: 1476-4687. DOI: 10.1038/nature04485

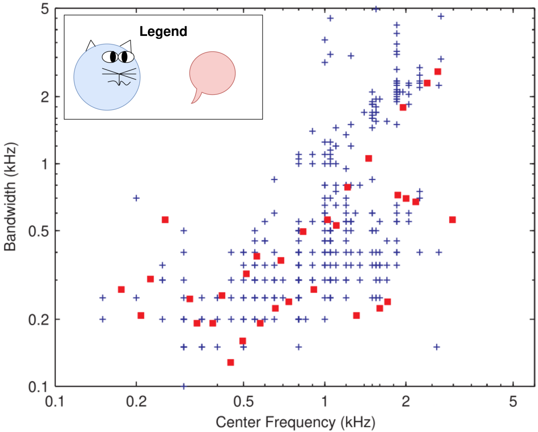
Background (2): Smith & Lewicki

- The functions below are the kernels that were found by Smith & Lewicki (about 5 to 15 ms in length)



Background (3): Smith & Lewicki

- It turns out that kernels trained on speech correlate extremely well with cat measurements!



- This suggests that the acoustic structure of speech is adapted to the auditory system.

Background (4a): So what does a representation look like?

The speech signal $s(t)$ is represented as a sum of N scaled (α_n) and shifted (τ_n) auditory kernels ϕ_n:

$$s(t) = \underbrace{\sum_{n=0}^{N-1} \alpha_n \phi_n(t - \tau_n)}_{\text{reconstruction}} + \underbrace{\varepsilon(t)}_{\text{error}}$$

Background (4b): So what does a representation look like?

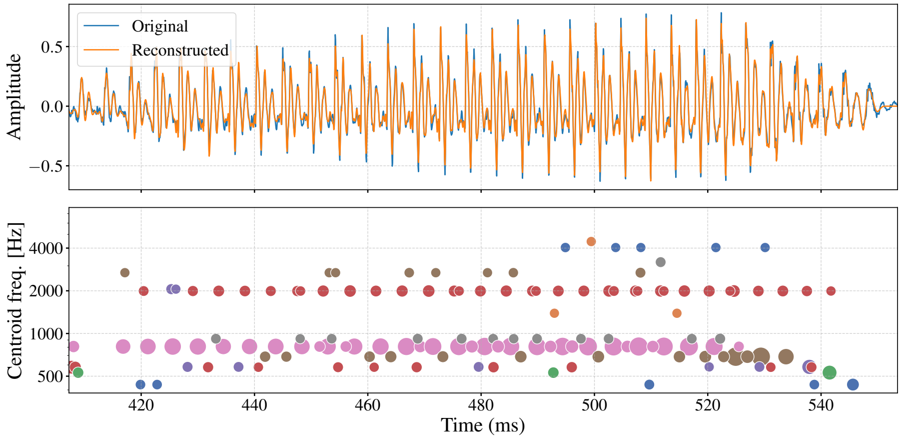


Figure: Speech segment of female speaker pronouncing /ae/ (as in "had"), with reconstruction and kernel activations. Point size represents $|\alpha_n|$, x-axis position τ_n, and y-axis the centroid frequency of kernel ϕ_n. Recall: $s(t) = \sum_{n=0}^{N-1} \alpha_n \phi_n(t - \tau_n) + \varepsilon(t)$

Some previous work (1): Auditory kernels across languages

We had a look at some spectral (frequency) properties of the kernels across languages. It turns out that they are very similar!

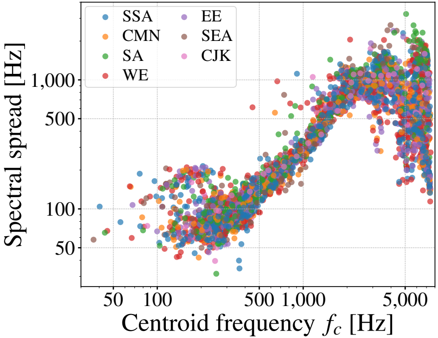


Figure: Two spectral properties (spread and centroid) of 32 kernels learned on 102 languages from 7 different geographic regions.

Some previous work (2): Auditory kernels for bat echolocation

A bachelor student of last year (Aleksandra Savova) had a look at a kernel-based representation for bat echolocation calls. She found that clustering based on the kernels used reveals distinct call shapes:

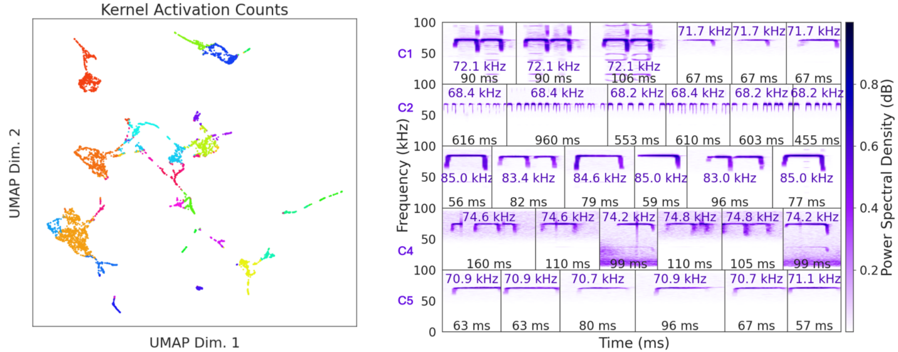


Figure: **Left:** clustering on kernel-based representation of bat echolocation calls. **Right**: Some example calls corresponding to clusters 1 to 5

Possible research questions

1. **RQ1:** Unifying auditory kernels and non-linear cochlear signal processing (Cascaded Envelope Interpolation).
2. **RQ2:** Effectively learning larger dictionaries.
3. **RQ3:** The effect of reverberation on the encoded speech signal
4. **RQ4:** Relation between kernels and speech production properties. For example: voiced vs unvoiced speech.
5. **RQ5:** Auditory kernels for different mammals. For example: bat social calls. (Expand on the results of Slide 10).
6. **RQ6 (not on project-forum):** Explaining the observed spectral-centroid spectral-spread curve (investigate the results of Slide 9).
7. **RQextra**: Extra topic on efficient coding to your wishes, for example: its relationship to speech command classification, comparing auditory kernels across different diversity dimensions (pathological speech, child speech, gender, different languages). Or your own research question after you have read the proposed literature (you would have to be quick if you want to propose your own).

Timeline (1): planning

- The planning of the bachelor research project is extremely tight.
- Have a good look at the planning on Brightspace!!!!!

Timeline (2): what to do till week 1?

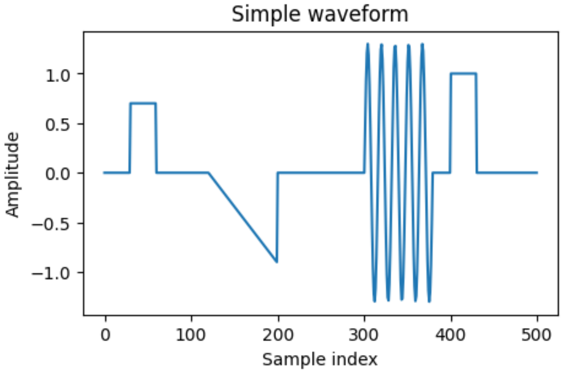
- **Select a research question**
- **Read the materials on the projectforum**. I will send a .zip with the papers ASAP.
- We will upload some materials and code to a GitLab page to get you started (at latest on the 9th of April)
- Feel free to send us questions you might have on the Teams! (We created a Teams channel: `RP-CS - 2026Q4 - Efficient Auditory Coding-PRJ`)
  - Or do you prefer Mattermost?
- Cluster access
- Faculty GitLab
- Lean Library PlugIn

Timeline (3): planning meetings

Plan our meetings

Extra: matching pursuit (1)

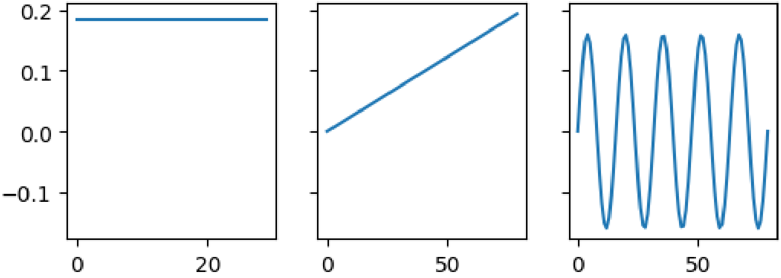
- Approach: iteratively select the dictionary element which best matches the residual of the waveform; This goes on until some stopping condition is reached.
- Consider the following waveform:



- What do you think would be good dictionary elements?

Extra: matching pursuit (2)

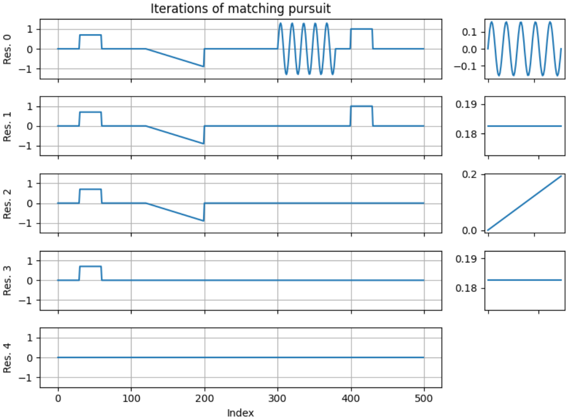
- What do you think would be good dictionary elements?
  1. A sawtooth
  2. A block
  3. A bit of a sinusoid



- We will iteratively select the element with the highest absolute cross-correlation.

Extra: matching pursuit (3)

- We iteratively select the element with the highest absolute cross-correlation and substract it from the waveform.



- The whole waveform is now represented with four tuples - `(amplitude, time index, element)`.

References I

Barlow, H. B. (Sept. 1961). "Possible Principles Underlying the Transformations of Sensory Messages". In: *Sensory Communication*. The MIT Press, pp. 216–234. DOI: 10.7551/mitpress/9780262518420.003.0013.

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